Ntuthuko Mdunge

ntuthukomdunge.is-a.dev
https://www.linkedin.com/in/ntuthuko-mdunge-4b8949258?utm_source=share_via&utm_content=profile&utm_medium=member_android

Personal Summary

- Motivated Electronic Engineering student at Durban University of Technology with hands on experience in embedded systems, circuit design, signal processing and applied machine learning. Eager to apply my technical skills to real world engineering challenges, keep learning from experienced professionals, and contribute to innovative, well built solutions.

Education

- **Durban University of Technology** 2027
  Enrolled in my BEng Tech in Electronics Engineering-NQF 7
  Still in progress.

Interests

- Chess
- Running
- Innovation
- Rugby

Experience

- **Durban University of Technology** 1 March 2026 - 1 June 2026
  Fundamentals of Power Engineering Tutor
  I assist students with the module content and taught them how to do the practicals and assisted them with their practicals.

Leadership Roles

- **Captain of the DUT chess team.(2025)**
- **Chairperson of the House Committee of DUT Alpine Residence. (2026)**
- **Team leader of the Space Science Centre Mars Rover Team.(2026)**
- **Team leader of SignalMyth SATNAC Team.(2026)**

Skills

- Problem Solving
- MATLAB
- Python programming
- Embedded Systems
- C Project building
- Machine Learning
- PCB Design
- Circuit Design.

Projects

- **SignalMyth: AI-Based LTE Coverage Prediction (SATNAC 2026 Industry Solutions Challenge, winner)**
  Developed a machine learning surrogate model that predicts LTE signal strength (RSRP) in seconds, compared with the 7–9 days Telkom's manual ray-tracing tool needs. The model used real SRTM terrain and ESA WorldCover land cover data for Durban and an XGBoost regressor. It achieved an R^2 of 0.958 and an RMSE of 2.63 dB against a physics based reference model, and was about 18x faster. The project won the competition and was presented as a technical report and pitch.
- **Mars Rover (DUT SSC Mars Rover Team, Cars4Mars African Rover Challenge 2026)**
  Contributed to the design and build of a rover for the Cars4Mars African Rover Challenge, including the bill of

materials, a dual architecture control system (Raspberry Pi 4 and ESP32), sensor suite planning and 3D-printed material selection (PLA for structural parts, TPU for tyres). I took part in hardware integration and testing and documented the build in a project strategy report and presentation.

- Pico Go: Line-Following and Obstacle-Avoiding Robot
  Designed and built an autonomous robot controlled by a Raspberry Pi Pico. It uses IR sensors, a TB6612 motor driver with two DC motors, an LCD, a buzzer and Bluetooth remote control. The project also included a custom 3D-modelled housing, system block and flow diagrams, and line tracking and remote control testing, and was presented to the faculty.
- Analogue IR Distance Sensor Circuit
  Designed and tested a microcontroller free proximity detector. A 555 timer in astable mode drives an IR transmitter, and an LM358 op-amp amplifies the reflected signal. Comparators then light green, orange or red LEDs to indicate distance. The circuit was verified in Multisim, then built on a breadboard and as a soldered PCB, and tested with an oscilloscope and multimeter on reflective and non reflective surfaces.
- Darkness-Activated Motion Detector
  Designed a circuit that switches on a light only when motion is detected in the dark, using an LDR, a PIR sensor, a 555 monostable timer and transistor driver stages. I simulated it in Multisim and laid out the PCB in Ultiboard with SMT and through hole components. I then built the prototype and tested it against the design requirements.

Achievements & Awards

- SATNAC 2026 Industry Solutions Challenge: Innovator of the Year (Telkom topic): Won Innovator of the Year with team SignalMyth at the SATNAC 2026 Industry Solutions Challenge. We built an AI model that predicts LTE coverage in seconds instead of the 7–9 days Telkom's manual ray tracing process takes. It was validated on real Durban terrain data ($R^2 = 0.958$, 2.63 dB RMSE), and I presented it as a technical report and pitch to industry judges.
- BRICS Competition of Skills Development and Technology Innovation 2025 (South Africa): Bronze Medal Won a team Bronze medal in the Technology Innovation category at the 2025 BRICS Competition of Skills Development and Technology Innovation (South Africa international leg). The project was Dr Kiosk, a medical self-service kiosk, and we represented Durban University of Technology. The certificate was issued on 28 August 2025 and is signed by the BRICS Business Council working group chairs.
- JCI Smart Cities Hackathon (Durban, May 2025): Second Place Placed second with team "The Lazy Company" at the JCI Smart Cities Hackathon during the JCI Africa and Middle East Conference in Durban (16–17 May 2025). We developed and pitched a smart city solution in a 24-hour competition format.

Languages

- English
- Zulu

Activities

- I am a competitive South African chess player and student athlete representing the Durban University of Technology (DUT) Chess Club. Registered under Chess South Africa (CHESSA), I regularly competes in national and university level tournaments, where I have built a strong reputation for sharp tactical execution and high pressure problem solving. Alongside my individual board performances, I leverages my strategic foresight to contribute to team preparation and analyze complex competitive scenarios.
- Active social runner who has completed four 10 km road races in Durban: the Ntenga Foundation Race, the Hollywood Bets Durban 10 km, the Absa Run Your City 10 km and the Garmin Run Series 10 km. Running regularly alongside my engineering studies and residence committee duties has built my discipline, consistency, time management and resilience, along with the habit of setting goals and working steadily towards them.

Additional Information

All my other projects and achievements can be found on my personal portfolio website, ntuthukomdunge.is-a.dev